

# WORKSHOP

## # Advanced Frontend Development: Ecommerce Website

### ## Project Overview

Build a fully responsive ecommerce website using HTML5, CSS3, and JavaScript. This project will demonstrate modern frontend development practices including semantic HTML, responsive design, and interactive JavaScript features.

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### ## Step 1: Semantic HTML5 Structure

#### ### 1.1 Basic HTML5 Document Setup

Create an `index.html` file with proper HTML5 structure:

```
``html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Ecommerce Store</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <!-- Content goes here →
</body>
</html>
``
```

#### ### 1.2 Semantic Layout

Define the main sections using appropriate semantic tags:

- `<header>`: For the navigation and main hero area with store branding
- `<section id="featured-products">`: Featured products section
- `<section id="categories">`: Product categories section
- `<section id="products">`: Main products listing
- `<section id="cart">`: Shopping cart section
- `<section id="checkout">`: Checkout form section
- `<footer>`: Footer with links and information

### ### 1.3 Navigation Structure

Create a navigation bar (`<nav>`) with links to:

- Home
- Products
- Categories
- Cart
- Contact

Include a mobile menu toggle button (hamburger icon) that is initially hidden on desktop.

### ### 1.4 Favicon

Add a favicon link:

```
```html
<link rel="icon" href="favicon.ico">
...
---
```

## ## Step 2: HTML5 Media Integration

### ### 2.1 Video Background

Embed a short, muted, looping video as a background for the hero section using the `<video>` tag:

- Set to `autoplay`
- Set to `loop`
- Set to `muted`

```
``html
<video autoplay loop muted id="video-background">
  <source src="video/background.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
...

```

### ### 2.2 Product Images

Add product images with proper `alt` attributes and dimensions:

```
``html

...

```

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## ## Step 3: CSS Styling and Responsive Design

### ### 3.1 Reset and Base Styles

Start with a CSS reset and base styles:

```
``css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

html {
  font-family: 'Poppins', sans-serif;
  background-color: #ffffff;
  color: #333333;
}

body {
  width: 100%;
  margin: 0;

```

```
padding: 2vw 5vw;  
}  
...
```

### ### 3.2 Desktop-First Approach

Style for desktop first, then use media queries to scale down:

#### **\*\*Desktop (Base Styles):\*\***

- Full navigation visible
- Multi-column product grid (3-4 columns)
- Large fonts and spacing
- Wide margins

### ### 3.3 Media Queries

#### **\*\*Tablet Breakpoint (max-width: 1023px):\*\***

- Reduce padding and margins
- 2-column product grid
- Slightly smaller fonts
- Adjusted navigation spacing

#### **\*\*Mobile Breakpoint (max-width: 767px):\*\***

- Mobile menu: navigation links hidden, hamburger icon shown
- Single column layouts for products
- Smaller fonts and padding
- Reduced spacing throughout
- Smaller product images

### ### 3.4 Component Styling

Style the following components:

- Product cards with hover effects
- Shopping cart items
- Buttons and form inputs
- Navigation menu
- Footer

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## ## Step 4: JavaScript Interactivity

### ### 4.1 Mobile Menu Toggle

Add a click event listener to the mobile menu button:

- Toggle a class (e.g., `.open``) on the navigation to make it visible
- Change the icon from hamburger (`≡`) to close (`×`)
- Update `aria-expanded`` attribute for accessibility

### ### 4.2 Shopping Cart Functionality

Implement cart features:

- Add products to cart
- Remove products from cart
- Update cart item quantities
- Calculate total price
- Display cart count badge

### ### 4.3 Scroll Indicator

Create a scroll progress bar:

- Listen for scroll events
- Calculate scroll position and total page height
- Dynamically update the width of a fixed progress bar element

### ### 4.4 Product Search/Filter

Implement search and filter functionality:

- Search products by name
- Filter by category
- Filter by price range
- Update product display dynamically

### ### 4.5 Form Validation

For checkout form:

- Add submit event listener
- Prevent default form submission
- Validate required fields (name, email, address, payment info)

- Display error messages next to invalid fields
- Show success message when form is valid

### ### 4.6 Product Image Gallery

If implementing product detail pages:

- Image carousel/slider
- Thumbnail navigation
- Zoom functionality

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## ## Step 5: Advanced Features (Optional)

### ### 5.1 Local Storage

- Save cart items to localStorage
- Persist cart across page refreshes
- Remember user preferences

### ### 5.2 Animations

- Smooth scroll behavior
- Fade-in animations for products
- Loading states
- Transition effects

### ### 5.3 Accessibility

- Proper ARIA labels
- Keyboard navigation support
- Focus management
- Screen reader compatibility

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## ## Implementation Checklist

### ### HTML Structure

- Semantic HTML5 layout with proper sections

- Navigation with mobile menu toggle
- Product listing structure
- Shopping cart structure
- Checkout form
- Footer with links
- Favicon link
- Video background in hero section

### ### CSS Styling

- Reset and base styles
- Desktop-first responsive design
- Tablet media query (max-width: 1023px)
- Mobile media query (max-width: 767px)
- Product card styling
- Cart styling
- Form styling
- Hover effects and transitions
- Scroll progress bar styling

### ### JavaScript Functionality

- Mobile menu toggle
- Shopping cart add/remove functionality
- Cart total calculation
- Scroll progress indicator
- Product search/filter
- Form validation
- Error message display
- Success message display

### ### Best Practices

- Clean, organized code
- Comments removed (production-ready)
- Consistent naming conventions
- Responsive images
- Optimized performance

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## ## File Structure

...

```
ecommerce-website/
├── index.html
├── css/
│   └── style.css
├── js/
│   └── script.js
├── images/
│   ├── product1.jpg
│   ├── product2.jpg
│   └── ...
├── video/
│   └── background.mp4
└── favicon.ico
```

...

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## ## Key Differences from Portfolio

### 1. **Content Focus:**

- Products instead of projects
- Shopping cart instead of contact form
- Categories instead of skills
- Product details instead of project descriptions

### 2. **Functionality:**

- Cart management (add/remove/update)
- Product filtering and search
- Checkout process
- Price calculations



### 3. **\*\*Layout:\*\***

- Product grid layouts
- Cart sidebar or modal
- Category navigation
- Product detail pages

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### **## Testing Checklist**

- Mobile menu works on small screens
- Products display correctly on all screen sizes
- Cart functionality works properly
- Form validation shows appropriate errors
- Scroll indicator updates correctly
- All links and buttons are functional
- Images load and display properly
- Video background plays correctly

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### **## Notes**

- Use desktop-first approach with media queries
- Ensure all JavaScript is wrapped in DOMContentLoaded
- Remove all comments from production code
- Test on multiple devices and browsers
- Follow semantic HTML5 best practices
- Maintain consistent styling throughout